

TatvaOps Omnichannel Lead Qualification & CRM System

[Aadhya AI Platform](#) | [Krsna Admin Panel](#)

AI-powered lead qualification across WhatsApp and Voice

Google Gemini intelligence | 11 service consultants

Full CRM for Presales, RM, and Admin teams

Integrated with Tatva Platform (withtatva.ai)

Version 1.0 | July 2026

TatvaOps Pvt. Ltd.

1. Executive Summary

The TatvaOps Omnichannel Lead Qualification & CRM System (codename: Aadhya) is an end-to-end platform that captures, qualifies, and manages homeowner leads across WhatsApp and Voice channels. AI consultants guide customers through structured questionnaires, generate project summaries, score leads, and sync data to the Tatva platform. The Krsna Admin Panel provides CRM capabilities for presales, relationship managers, and administrators.

Key Capabilities

- Dual-channel intake: WhatsApp (Twilio) and Voice (Vapi + ElevenLabs)
- 11 specialized AI consultants for TatvaOps home services
- Hybrid qualification: MCQ buttons, free-text, and file uploads
- Google Gemini-powered conversation, extraction, and summaries
- Lead scoring: Hot / Warm / Cold tiers
- CRM: lead assignment, progress tracking, meet scheduling, sales targets
- Deep integration with Tatva Platform API (withtatva.ai)

2. System Architecture

End-to-end data flow:

- 1. Customer contacts via WhatsApp or Voice
- 2. EVA receptionist greets and presents service menu
- 3. Customer selects one of 11 TatvaOps services
- 4. Specialized AI consultant runs hybrid qualification flow
- 5. ConversationController + Gemini extract structured enquiry data
- 6. Lead scored, summary generated, enquiry submitted to Tatva API
- 7. Data persisted in Redis (sessions) and Supabase (CRM/enquiries)
- 8. Krsna CRM panel: assign, track, and close leads

Deployment Architecture

Component	Technology	URL / Host
Backend API	FastAPI on Render	tatvaops-omni-dev.onrender.com
Admin UI (CRM)	React + Vite on Vercel	Vercel deployment /krsna
Session Store	Upstash Redis	REST API, 24h TTL
Persistent DB	Supabase (PostgreSQL)	Enquiries, CRM, files
File CDN	CloudFront	Attachment URLs
Tatva Platform	withtatva.ai API	Users, presales, vendors

3. Technology Stack

Backend

Technology	Purpose
Python 3.11+	Runtime
FastAPI + Uvicorn	REST API, webhooks, admin APIs
Pydantic	Configuration and data schemas
Google Gemini 2.0 Flash	AI conversation, extraction, summaries
Twilio	WhatsApp messaging and interactive templates
Vapi + Deepgram	Voice calls and transcription
ElevenLabs	Voice synthesis
Upstash Redis	Live session state
Supabase	PostgreSQL + file storage
httpx	External API client

Frontend (Krsna Admin Panel)

Technology	Purpose
React 18 + TypeScript	Single-page application
Vite	Dev server and production build
Tailwind CSS	UI styling
React Router	Client-side routing
Recharts	Dashboard charts and analytics
Lucide React	Icon library
date-fns	Date formatting

4. Services & AI Consultants

Each TatvaOps service has a dedicated AI consultant persona with a service-specific qualification flow defined in JSON configuration files.

#	Service	Consultant
1	Residential Construction	Aravind Narayanan
2	Home Interiors	Aadhya
3	Painting & Waterproofing	Manjunath Gowda
4	Electrical Services	Vivek Shetty
5	Plumbing Services	Suresh Kumar
6	Solar Rooftop	Kavya Nair
7	Event Management	Meera Iyer
8	Property Development	Vikram Desai

#	Service	Consultant
9	Home Automation	Riya Mehta
10	Farm Infrastructure Setup	Anil Reddy
11	Irrigation Automation	Deepak Patil

5. CRM Features (Krsna Admin Panel)

User Roles

Role	Access Level
admin	Full: dashboard, presales, users, vendor leads, vendors, system health
presales	Team dashboard, My Leads, My Projects
rm (Relationship Manager)	Team dashboard, My Leads, My Projects

CRM Pages

Page	Path	Description
Login	/krsna	CRM team authentication (email + password)
Dashboard	/krsna/dashboard	Stats, charts, lead acquisition, team performance
Pre-sales	/krsna/presales	Admin view: assign leads to staff/vendors
My Leads	/krsna/my-leads	Staff assigned leads, comments, meet scheduling
My Projects	/krsna/my-projects	Employee projects from Tatva API
Users	/krsna/users	Manage CRM team users (admin only)
Vendor Leads	/krsna/vendor-leads	Vendor-sourced leads (admin only)
Vendors	/krsna/vendors	Approved vendors from Tatva (admin only)
System Health	/krsna/system	API health monitoring (admin only)

Lead Management

- Assign leads to presales staff, RM, Tatva employees, or vendors
- Custom progress stages and comment logs per lead
- Google Meet scheduling via Tatva meet-links API
- Sales targets per staff (day/month/quarter/year) with achievement %
- Lead scoring: Hot (75+), Warm (45-74), Cold (<45)
- UTM and lead acquisition filtering on dashboard

Lead Status Workflow

unassigned -> assigned -> in_progress -> presales_completed

6. External Integrations (Tatva Platform)

The system integrates with the Tatva main API (api.withtatva.ai / devopsapi.withtatva.ai):

Integration	Purpose
User check / register	Phone lookup on first message; register new users
Service questionnaire	POST completed enquiry to Tatva
Presales API	Submit lead after create-project decision

Integration	Purpose
Presales / Vendor leads	Fetch leads for CRM pages
Employees API	Sales team list for assignment
Employee projects	RM/presales My Projects view
Meet links	Schedule customer meetings
User addresses	Returning user saved locations

7. Data Storage

Store	Contents	Fallback
Upstash Redis	Live chat sessions, conversation state	In-memory (data lost on restart)
Supabase PostgreSQL	enquiries, summaries, sessions_log, CRM tables	Not available without config
Supabase Storage	Uploaded files (enquiry-files bucket)	N/A
CloudFront CDN	Public attachment URLs	N/A

Key Database Tables

Table	Purpose
enquiries	Structured enquiry field data
project_summaries	AI-generated project summaries
sessions_log	Conversation session logs
enquiry_attachments	File upload metadata
crm_users	CRM team accounts (admin/presales/rm)
lead_assignments	Lead ownership and status tracking
sales_targets	Staff performance targets

8. Project Structure

omnichannel-main/	
backend/	
main.py	FastAPI entry, mounts /krsna UI
config.py	Environment configuration
agents/chat/	WhatsApp (Twilio) webhook handler
agents/voice/	Voice (Vapi) webhook handler
intelligence/	AI orchestration, flows, scoring
integrations/	Tatva API clients
crm/	CRM users and lead assignments
admin/	Admin and CRM API routers
summarizer/	AI summary generation
storage/	Redis + Supabase persistence
admin-ui/	React CRM dashboard (Krsna)
scripts/	SQL schemas, Twilio templates, utilities
tests/	Pytest test suite
render.yaml	Render deployment config
.env	Environment variables

9. Key Environment Variables

Variable	Purpose	Required
GEMINI_API_KEY	Google Gemini AI	Yes

Variable	Purpose	Required
TWILIO_*	WhatsApp messaging and templates	For WhatsApp
VAPI_API_KEY	Voice calls	For Voice
ELEVENLABS_*	Voice synthesis	For Voice
UPSTASH_REDIS_*	Session storage	Recommended
SUPABASE_URL + KEY	Database and CRM	Recommended
ADMIN_PASSWORD / API_KEY	Legacy admin auth	Yes
TATVA_USERS_API_BASE_URL	Tatva platform API	Yes
BASE_URL	Public URL for webhooks	Production
CORS_ORIGINS	Allowed frontend origins	Production

10. Local Development Setup

Prerequisites: Python 3.11+, Node.js 18+, Gemini API key

Backend:

```
pip install -r requirements.txt
cp .env.example .env    (configure API keys)
uvicorn backend.main:app --reload --port 8000
```

Admin UI:

```
cd admin-ui && npm install && npm run dev
Open http://localhost:3000/krsna
```

API Documentation: <http://localhost:8000/docs>

Health Check: <http://localhost:8000/health>

11. Lead Scoring Model

Leads are automatically scored (0-100) and tiered based on qualification completeness:

Factor	Points
Field completion percentage	Up to 50 points
AI summary generated	+20 points
File attachments uploaded	+15 points
Budget range provided	+10 points
Timeline provided	+5 points
Urgent timeline (ASAP, this month)	+10 points

Tiers

Tier	Score Range	Meaning
Hot	75 - 100	High-intent, well-qualified lead
Warm	45 - 74	Moderate intent, partial qualification
Cold	0 - 44	Low intent or incomplete data

12. Key API Endpoints

Endpoint	Method	Description
/webhook/whatsapp	POST	Twilio WhatsApp webhook
/webhook/vapi	POST	Vapi voice webhook
/admin/crm-login	POST	CRM team login
/admin/dashboard	GET	Dashboard statistics
/admin/presales	GET	Presales lead list
/admin/lead-assignments/*	PATCH	Assign and update leads
/admin/team-dashboard	GET	Team performance metrics
/health	GET	Service health check
/docs	GET	Swagger API documentation

This document describes the TatvaOps Omnichannel Lead Qualification & CRM System (Aadhya / Krsna). For technical support or updates, contact the TatvaOps engineering team.